

Homework Problems. Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.

1. Let $(\mathbb{Q}, +)$ be the group of rational numbers under addition.

- a. Is $\mathbb{Z}$ a subgroup of $(\mathbb{Q}, +)$?

Proof. Yes. By the subgroup test: $\mathbb{Z} \neq \emptyset$ (since $1 \in \mathbb{Z}$) and $\mathbb{Z} \subseteq \mathbb{Q}$ (since $a \in \mathbb{Z} \Rightarrow \frac{a}{1} \in \mathbb{Q}$). If $a, b \in \mathbb{Z}$ then $b^{-1} = -b$ so $ab^{-1} = a - b \in \mathbb{Z}$.

Alternatively, simply note that $\mathbb{Z} \subseteq \mathbb{Q}$ and that $(\mathbb{Z}, +)$ is a group by a previous exercise. $\square$

- b. Is $\mathbb{Z}^+ = \{0, 1, 2, 3, \dots\}$ a subgroup of $(\mathbb{Q}, +)$?

Proof. No. If $a \in \mathbb{Z}^+$ then $aa^{-1} = 0$ implies that $a^{-1} = -a$, and $-a \notin \mathbb{Z}$ if $a \geq 0$ (groups/subgroups must contain the inverse of each element). $\square$

2. Determine if the following are subgroups.

- a. Is $\mathbb{Q}$ a subgroup of the group, $(\mathbb{C}, +)$, the complex numbers under addition?

Proof. Yes. By the subgroup test: If $\frac{a}{b} \in \mathbb{Q}$ for $a, b \in \mathbb{Z}$ then $\frac{a}{b} = \frac{a}{b} + 0 \cdot i \in \mathbb{C}$, and $\frac{1}{1} \in \mathbb{Q} \Rightarrow \mathbb{Q} \neq \emptyset$. If $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ for some $a, b, c, d \in \mathbb{Z}$ then $\left(\frac{c}{d}\right)^{-1} = -\frac{c}{d}$, and $\frac{a}{b} - \frac{c}{d} = \frac{ad - cb}{bd} \in \mathbb{Q}$ since $ad - cb$ and $bd \in \mathbb{Z}$. $\square$

- b. Is $\mathbb{Z} - \{0\}$ a subgroup of the group, $(\mathbb{Q} - \{0\}, \cdot)$, the non-zero rational numbers under multiplication?

Proof. No. $e = 1$ so if $a \in \mathbb{Z}$ and $a \neq 1$ then $a \cdot a^{-1} = 1 \Rightarrow a^{-1} = \frac{1}{a} \notin \mathbb{Z}$. $\square$

3. Prove that the intersection of two subgroups H and K of a group G is a subgroup.

Proof. Let $H, K \leq G$ and $a, b \in H \cap K$. By the subgroup test: Since H and K are subgroups $e \in H$ and $e \in K \Rightarrow e \in H \cap K$ where e is the identity of G . Thus $H \cap K \neq \emptyset$. Also $H \cap K \subseteq G$. $b \in H$ implies that $b^{-1} \in H$ since H is a subgroup. So then $ab^{-1} \in H$ since $a \in H$ and both H is closed. Similarly $ab^{-1} \in K$ so then $ab^{-1} \in H \cap K$, and $H \cap K \leq G$. $\square$

4. Let $H_1, H_2, \dots$ be subgroups of G . Prove that $H_1 \cap H_2 \cap \dots$ is a subgroup of G .

Proof. $H_1 \cap H_2 \cap \dots = \cap_{i=1}^{\infty} H_i$. By the subgroup test: $e \in \cap_{i=1}^{\infty} H_i$ and each $H_i \leq G$ so then $\cap_{i=1}^{\infty} H_i \neq \emptyset$ and $\cap_{i=1}^{\infty} H_i \subseteq G$. If $a, b \in \cap_{i=1}^{\infty} H_i$ then $b^{-1} \in H_i$ for each i since every H_i is a subgroup. So then $ab^{-1} \in H_i$ since each H_i is closed which implies that $ab^{-1} \in \cap_{i=1}^{\infty} H_i$. $\square$

5. Recall that $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ is a group under modular addition.

- a. Show that $H = \{0, 2, 4\}$ and $K = \{0, 3\}$ are subgroups of $\mathbb{Z}_6$.

Proof. $H = \langle 2 \rangle$ and $K = \langle 3 \rangle$, i.e., they are both cyclic. So they are both subgroups by theorem stated in class. (**Thm:** If G is a subgroup and $a \in G$ then $\langle a \rangle \leq G$ under the operation of G). $\square$

- b. Show that $H \cup K$ is not a subgroup of $\mathbb{Z}_6$.

Proof. $H \cup K = \{0, 2, 3, 4\}$. $2, 3 \in H \cup K$, but $2 + 3 = 5 \notin H \cup K$. $\square$

- c. Is the union of two subgroups of G necessarily a subgroup G ?

Proof. No. H and K are both subgroups by part (a), but $H \cup K$ is not a subgroup by part (b). $\square$

6. Let G be an abelian group and H be a subgroup of G . Prove that $S(H) = \{x|x \in G \text{ and } x \cdot x \in H\}$ is also a subgroup of G .

Proof. By definition, $S(H) \subseteq G$. Also the identity is in every subgroup so $e \in H$. Thus $e \cdot e \in S(H)$, and $S(H) \neq \emptyset$. If $x, y \in S(H)$ then $x \cdot x \in H$ and $y \cdot y \in H \Rightarrow y^{-1}y^{-1} \in H$ since H is a subgroup. So then $(x \cdot x)(y^{-1}y^{-1}) = xy^{-1}xy^{-1}$ since G is abelian. Thus $xy^{-1} \in S(H)$, and $S(H) \leq G$. $\square$

7. Let G be a group. Prove that G has a subgroup with one element. This group is called the *trivial group*.

Proof. Because G is a group it has an identity, e . $\{e\} \subseteq G$, $\{e\} \neq \emptyset$, and $e \cdot e^{-1} = e \cdot e = e \in \{e\}$. Thus $\{e\} \leq G$. (Alternatively, note that $\{e\}$ is cyclic and use the theorem as in problem 5.a). $\square$

Definition. Let G be a group. If there is a single $g \in G$ such that every element in G is equal to g^n for some $n \in \mathbb{N}$ then G is a *cyclic group* and x is called the *generator* of G .

8. Prove that:

- a. $\mathbb{Z}_n$ is a cyclic group under modular addition for any $n \in \mathbb{Z}$ such that $n > 0$. (you can use the fact that it is a group).

Proof. $n > 0$, so then $1 \in \mathbb{Z}_n$, and $\langle 1 \rangle = \{1, 2, 3, \dots, n-1, 0\} = \mathbb{Z}_n$. Thus $\mathbb{Z}_n$ is cyclic generated by 1. $\square$

- b. $\mathbb{Z}$ is a cyclic group.

Proof. We have already shown that $\mathbb{Z}$ is a group. $1 \in \mathbb{Z}$, and $\langle 1 \rangle = \{\dots, -2, -1, 0, 1, 2, \dots\}$. ($1^{-2} = -2$; As stated in class, all cyclic groups of the same order are *isomorphic*. So $\mathbb{Z}$ is the only infinite cyclic group up to isomorphism). $\square$

9. Find all generators of

- a. $\mathbb{Z}_6$, $\mathbb{Z}_8$, and $\mathbb{Z}_{10}$ (note: $\mathbb{Z}_n$ is often notated $\mathbb{Z}_n$).

$\mathbb{Z}_6$ $\langle 1 \rangle = \{1, 2, 3, 4, 5, 0\}$ (note: 1 is a generator of $\mathbb{Z}_n$ for any n) and $\langle 5 \rangle = \{5, 4, 3, 2, 1, 0\}$ so $\langle 1 \rangle = \langle 5 \rangle = \mathbb{Z}_6$.

$\mathbb{Z}_8$ $\langle 1 \rangle, \langle 3 \rangle = \{3, 6, 1, 4, 7, 2, 5, 0\}$, $\langle 5 \rangle = \{5, 2, 7, 4, 1, 6, 3, 0\}$, $\langle 7 \rangle = \{7, 6, 5, 4, 3, 2, 1, 0\}$.

$\mathbb{Z}_{10}$ Similarly to above we find that the generators are: $\langle 1 \rangle, \langle 3 \rangle, \langle 7 \rangle$, and $\langle 9 \rangle$

Definition. Let $X = \{1, 2, \dots, n\}$. The *symmetric group on X* , S_n , is the collection of all bijections $\sigma : X \rightarrow X$. That is, it is the set of all functions that rearrange the elements of X . (S_n is also called the *permutation group of the set X*).

10. Let S_3 be the symmetric group of permutations on three elements. Show that S_3 is:

- a. Non-abelian.

Proof. Let $\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ and $\sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$. $\sigma_1\sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = \sigma_2\sigma_1$. $\square$

- b. Has a cyclic subgroup of order 3.

Proof. $\sigma_1^1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$, $\sigma_1^2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$, and $\sigma_1^3 = \sigma_1^2\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = e$. So then $|\sigma_1| = 3$, and we know that $\langle \sigma_1 \rangle \leq S_3$. $\square$